Illusory Edge: Persistent Label Errors, Disagreement Audits, and the Limits of Measured Predictability

Johan Abraham

***Abstract*—A forecasting model's measured skill can exceed its true skill when the labels it is scored on contain errors that persist within entities. We give an exact identity for the effect of label errors on the measured accuracy difference between two fixed decision rules, show that entity-persistent errors can reverse the sign of an edge without being detected by time-ordered validation, and derive a disagreement audit: a finite-population bound on the true edge from hand-checking a random sample of the rows on which two rules disagree. In a reproduced earnings-guidance model, rows later removed as mislabelled carried an apparent edge of $+20.1$ points; on a fixed cohort of 587 correctly labelled forecasts the edge was indistinguishable from zero before and after retraining, and an audit of 40 disagreement rows bounds the true edge below by -6.10 points, so no positive edge can be certified. We separate the Bayes ceiling of a stated information set from algorithm-specific learnability and report achieved skill relative to point-in-time base rates in eleven tasks. In exploratory studies of two low-dimensional chaotic systems, the samples a nearest-neighbour learner needs to come within 0.05 of an ensemble reference grow steeply with the horizon (close to exponentially for Lorenz-63), at rates that depend on the learner and the tolerance; for Lorenz-96 the rates are mostly not identifiable within 10^6 samples. On IBM processors, next-calibration readout and two-qubit errors are partly forecastable and coherence times at most weakly. Simulated quantum classifiers showed no detectable advantage on classical data.**

***Index Terms*—Bayes error, forecast verification, label noise, data auditing, nearest-neighbour classification, chaos, quantum hardware.**

I. Introduction

ACCURACY against a coin is a poor guide to whether an outcome is predictable; the relevant reference is the base rate, and the relevant question is how much of the outcome's uncertainty a stated information set can remove. A second, less familiar hazard concerns the labels themselves. When label errors are attached to particular entities and persist over time, a flexible model can learn them; because future labels share the same errors, time-ordered validation rewards the model for having done so. This paper analyses that mechanism, gives a practical audit for it, and documents it in a production forecasting model.

The work began with a model that forecasts whether a firm's diluted GAAP earnings exceed the midpoint of management's original guidance [1]. An early version showed an edge over the issuer base rate that disappeared when label-extraction errors were removed; the errors were concentrated in particular issuers (guidance for continuing operations scored against total earnings, quarter-mapping errors) and repeated over time.

J. Abraham is an independent researcher. Code, archived predictions, the preregistration and derived results are available for reproduction (Appendix B).

A. Contributions and their status

We classify each item as established (E), synthesis (S), new proposition (N) or new empirical finding (F), following a prior-art audit (Section II).

1) (N, central) An *inflation identity* for the effect of label errors on the measured accuracy difference between fixed rules, a sign-reversal construction for entity-persistent errors, and the conditions under which time-ordered validation cannot detect them (Section IV). The phenomenon is a specialisation of instance-dependent label noise [2], [3]; the identity and its validation-blindness conditions are, to our knowledge, not stated elsewhere.
2) (N) A *disagreement audit* with an exact finite-population (hypergeometric) guarantee and a weighted worst-case bound, extending the discordant-pair logic of McNemar's test and related to active label cleaning [4] and active testing [5].
3) (F) A fixed-cohort reanalysis of the earnings model that separates sample selection, training-label correction and evaluation-label correction, and an audit of its disagreement rows.
4) (S) A framework separating the Bayes ceiling of an information set from algorithm-specific learnability; the knowable fraction is Theil's uncertainty coefficient [6], and log-loss skill decompositions follow [7], [8].
5) (F, exploratory) Achieved skill in eleven tasks with time-block uncertainty; learnability–horizon curves for two nearest-neighbour learners in five chaotic systems (rates identifiable in two); forecastability of IBM processor calibrations; a live CHSH experiment and a reanalysis of a loophole-free Bell test.

The chaos and hardware studies are exploratory and could each support a separate paper; here they serve as controlled and physical illustrations of the framework.

II. Related Work

Label noise. Learning with class- or instance-dependent noise is surveyed in [2], [9]; part-dependent noise models errors that depend on input features [3]; noise-robust losses [10] and confident learning [11] address training under such noise. Errors in widely used test sets can change model rankings [12], models exploit dataset artefacts and shortcuts [13], and leakage undermines machine-learning-based science [14]. We study the evaluation consequence of errors that are constant within entities, and an audit restricted to disagreement rows. Prioritising labels for re-checking is studied as active label cleaning [4], and label-efficient estimation of test performance as active testing [5].

Predictability and information. The uncertainty coefficient [6] normalises mutual information by outcome entropy; information-theoretic forecast scores and their decompositions are developed in [7], [8], [15]. Limits of predictability have been studied in human mobility [16], social systems [17]–[19] and weather [20]–[22].

Nearest neighbours and chaos. 1-NN converges to error $\mathbb{E}[2r(1-r)]$ [23], and k-NN with $k \to \infty$, $k/n \to 0$ is consistent [24]. Nearest-neighbour and local forecasting of chaotic systems is classical [25], [26]; data-driven forecasting with reservoir computers is reviewed in [27]. Our chaos study measures sample requirements of named learners against an ensemble reference.

Quantum. The bound on accuracy from classical inputs is a corollary of data processing; the operational meaning of min-entropy [28], CHSH and Tsirelson's bound [29], [30], loophole-free Bell tests and their statistics [31]–[33], and drift and relaxation-time fluctuations in superconducting qubits [34]–[37] provide the context.

III. Framework

Formal definitions, assumptions and proofs are collected in Appendix A; we summarise them here.

Information sets and ceilings. A forecasting task specifies an evaluation distribution P of a test example (X, Y), where X collects everything available at the decision time; we assume $0 < P(Y = 1) < 1$. The Bayes ceiling of this information set is $A^*(\mathcal{F}) = \mathbb{E}\max\{\eta(X), 1-\eta(X)\}$ with $\eta(x) = P(Y = 1 \mid X = x)$. We use "Bayes ceiling relative to $\mathcal{F}$" rather than "physical ceiling": it is a property of the stated information and distribution, not of nature. No decision rule using only X exceeds it (Lemma 1).

Learnability. For a named learning algorithm $\mathcal{A}$ trained on n examples independent of the test example, $A_n(\mathcal{A}) = \mathbb{E}_{S_n} \Pr\{\mathcal{A}(S_n)(X) = Y\}$ and $G_n(\mathcal{A}) = A^*(\mathcal{F}) - A_n(\mathcal{A})$. All finite-sample statements below concern specific algorithms; we make no claim about an optimal learner.

Knowable fraction and achieved skill. $\kappa(\mathcal{F}) = I(Y;X)/H(Y)$ is a population quantity. What we observe is the log-loss skill of a fitted forecast $\hat{p}$ relative to a point-in-time base-rate forecast p_0, $\mathrm{LSS} = 1 - \bar{\ell}(\hat{p})/\bar{\ell}(p_0)$. In expectation,

$$\mathrm{LSS} = 1 - \frac{H(Y \mid X) + \mathbb{E}\,\mathrm{KL}(\eta \| \hat{p})}{H(Y \mid \mathcal{F}_0) + \mathbb{E}\,\mathrm{KL}(\eta_0 \| p_0)}, \tag{1}$$

so achieved skill combines the information gap between $\mathcal{F}_0$ and $\mathcal{F}$ with the misspecification, estimation, calibration and drift losses of both forecasts. A small LSS is therefore not evidence of a small Bayes ceiling, and differences between nested fitted models are achieved score improvements, not conditional mutual informations; they can be negative.

Closed forms used as references. For a qubit $R_y(\theta)|0\rangle$ with $\theta \sim U(0,\pi)$ known, $A^* = \frac{1}{2} + \frac{1}{\pi}$ and $\kappa = 1/\ln 2 - 1$; with readout errors e_0, e_1 the Bayes ceiling is given in Proposition 6 and equals $\frac{1}{2} + (1 - e_0 - e_1)/\pi$ only when $e_0 = e_1$. For exchangeable entities with Beta-distributed rates, the oracle and finite-history Bayes accuracies are given in Proposition 7; on real data, where hyperparameters are estimated, these are model-based approximations.

Nearest-neighbour diagnostics. Because 1-NN converges to $\mathbb{E}[2r(1-r)]$ with $r = \min(\eta, 1-\eta)$, its asymptotic excess error is $\mathbb{E}[r(1-2r)]$, which is positive whenever labels remain uncertain given X. Inverting the Cover–Hart inequality with a finite-sample error estimate, or combining it with the Hellman–Raviv bound, yields only asymptotic plug-in diagnostics; we report them as such and never as certified bounds.

IV. Illusory Edge and Disagreement Audits

Let $\tilde{Y}$ be the recorded label, $q(x) = P(Y = 1 \mid X = x)$ and $\tilde{q}(x) = P(\tilde{Y} = 1 \mid X = x)$. Statements concern fixed decision rules f, g unless stated.

Proposition 1 (Inflation identity)**.** *For fixed rules, the measured edge $\tilde{\Delta}$ and the true edge Δ satisfy $\tilde{\Delta} - \Delta = 2\,\mathbb{E}[(f(X) - g(X))(\tilde{q}(X) - q(X))]$.*

Measured edge is inflated exactly when f departs from g in the direction of the label distortion. Two consequences follow (proofs in Appendix A).

Proposition 2 (Symmetric independent noise)**.** *If labels flip with probability $\rho < \frac{1}{2}$ independently of (X, Y), then $\tilde{\Delta} = (1-2\rho)\Delta$ for fixed rules, and the Bayes rule for the recorded labels coincides with the clean Bayes rule.*

Proposition 3 (Sign reversal)**.** *If $q \equiv m > \frac{1}{2}$, a share ω of entities (by outcomes) has $\tilde{q} = 0$ with $(1-\omega)m > \frac{1}{2}$, $g \equiv 1$, and f is the entity-wise Bayes rule for the recorded labels, then $\tilde{\Delta} = \omega > 0$ while $\Delta = -\omega(2m-1) < 0$.*

Proposition 4 (Time-ordered validation)**.** *If within each entity the outcomes and corruption are i.i.d. over time with $\tilde{q}_i \neq \frac{1}{2}$, each evaluated entity has an unbounded history, and the entity rule is the majority of past recorded labels, then walk-forward decisions converge almost surely to $\mathbf{1}\{\tilde{q}_i > \frac{1}{2}\}$ and the walk-forward measured edge converges to the in-sample measured edge. With a pooled base-rate fallback for held-out entities, the leave-entity-out entity-specific edge is zero.*

The leave-entity-out check therefore flags entity-specific edge but cannot distinguish genuine persistence from persistent noise.

Proposition 5 (Disagreement audit)**.** *On a fixed evaluation population with weights w_j and disagreement set D (N_D rows), only wrong labels in D affect the measured edge, and $|\tilde{\Delta} - \Delta| \le 2\sum_{j \in D_{\mathrm{wrong}}} w_j / \sum_j w_j$. If n rows sampled uniformly without replacement from D contain e errors and K_U is the exact one-sided $(1-\alpha)$ hypergeometric upper bound on $|D_{\mathrm{wrong}}|$, then with probability at least $1-\alpha$, $\Delta \ge \tilde{\Delta} - 2W_{(K_U)}/\sum_j w_j$, where $W_{(K)}$ is the sum of the K largest weights in D.*

The bound concerns the measured edge on the fixed population as labelled, not future predictive edge. The binomial Clopper–Pearson bound [38] is valid for sampling with replacement and conservative here.

V. Experimental Design

Tasks. The real-data tasks are measurable rain at 61 stations (with a numerical-weather-prediction, NWP, benchmark over 17 months), football home wins (with closing odds), contested U.S. House seats and presidential state outcomes, earnings guidance, next-day stock direction for current large U.S. stocks and funds, weekly earthquake occurrence in 2° cells, and IBM processor calibrations. Controls are a qubit with known ceiling and five chaotic systems. A feature-and-timestamp specification for every task, with the leakage audit, is part of the reproducibility package; no demonstrated leakage was found, and the unresolved risks (revised catalogues and observations, survivorship in the stock universe, guidance revisions, polling delays) are listed in Section VIII.

Protocol. Every real-data task is evaluated walk-forward and point in time: a row is scored only by models fitted on labels available before its decision time, and all preprocessing (winsorisation, standardisation, priors, PCA) is fitted on training data only. Models are the point-in-time base rate, an empirical-Bayes entity model [39], [40], L_2 logistic regression, gradient boosting and nested variants that include the base-rate forecast as an input.

Inference. Intervals use entity-clustered bootstraps and, where shocks are shared across entities, time-block bootstraps (calendar month, election year or quarter). Quantum–classical comparisons select each family's best model on a held-out validation split and test once; we report entity-clustered and month-block bootstrap intervals for the log-loss difference and a one-sided Diebold–Mariano test [41] on the row-level loss differential ordered by decision time (Newey–West variance, lag $\lfloor T^{1/3} \rfloor$), and with one-sided bootstrap p-values over entities and over calendar months; Holm's correction [42] across the eight domains is applied to the larger of the two bootstrap p-values. Panel dependence makes the Diebold–Mariano test approximate; the bootstraps are primary.

Preregistration. A document fixing hypotheses H1–H6, primary models and predicted skill ranges was hashed (SHA-256 prefix `abee64cc`) and committed to a local repository before data for the new real-data tasks were downloaded. The hash identifies the content; the local commit time is not independently verifiable and the hash was not deposited with a third-party registry, so the timing claim rests on our records. Preregistered analyses are preserved; deviations (1-NN diagnostic inputs, dropping tennis, nested models, refitting the Lorenz-63 scaling with 1-NN) and all analyses added in revision (multi-system chaos, hardware, Bell data, fixed-cohort earnings, time-block intervals, validation-based model selection) are exploratory. The earnings design predates the preregistration.

VI. Results

A. Illusory edge in simulation and in a production model

In 9,360 simulated panels, symmetric noise left measured and true edges nearly equal (means +0.0261 and +0.0288); under persistent corruption satisfying the hypotheses of Proposition 3, the simulation reproduced the predicted reversal at finite history (for $m = 0.75$, $\omega = 0.2$: measured +0.190, true −0.114; theory +0.200 and −0.100). These are numerical checks of the propositions, not independent evidence.

TABLE I
Earnings Guidance: Mechanisms Behind the Apparent Edge

Quantity (EB edge over base rate, points)	Value [95% CI]
Removed rows (n = 76), archived predictions, original labels	+20.1
Fixed cohort (n = 587), original training history	+0.46 [−3.63, +4.70]
Fixed cohort, retrained on cleaned history	−0.81 [−3.92, +2.22]
All new-ticker rows before cleaning	+2.88
contribution of later-removed rows	+2.48 [+0.70, +4.70]
All new-ticker rows after cleaning	−0.65
Audit: errors / verified rows (1 unverifiable)	2 / 39
Audit lower bound, hypergeometric, weighted (in parentheses: unverifiable row counted as wrong)	−6.10 (−7.14)

The production model reproduced byte for byte from its archived code and data; this establishes consistency with the archive, not correctness. We separate three mechanisms on the 98 new tickers (Table I). *Sample selection:* the 76 scored rows that the cleaning rules removed carried an empirical-Bayes edge over the base rate of +20.1 points when the archived predictions are scored against their original labels; these rows were excluded, not relabelled, and corrected labels for them (for example, total-company guidance where only continuing-operations guidance was published) are not available. *Training-label correction:* on the fixed cohort of 587 forecasts that were correctly labelled and scored in both runs, the edge was +0.46 points (−3.63 to +4.70) with the original training history and −0.81 (−3.92 to +2.22) after retraining on the cleaned history, so correcting the training labels changed little on this cohort. *Aggregate:* the pooled edge fell from +2.88 to −0.65 points, with the later-removed rows contributing +2.48 (+0.70 to +4.70). The removed rows' recorded outcomes beat the midpoint in 26% (continuing-operations rows) and 22% (quarter-mapping errors) of cases, against 59% for correct rows, consistent with persistent, learnable misses.

Audit. We drew 40 of the 82 harvested disagreement rows of the cleaned combined universe uniformly without replacement (fixed seed) and checked each against the press release; 39 were verifiable and 2 were wrong (continuing-operations guidance missed by the cleaning rule). The exact hypergeometric bound gives at most 10 wrong rows in D (rate 0.122, against 0.153 from Clopper–Pearson); the weighted worst case places the true edge above −6.10 points (measured −0.87). Counting the unverifiable row as wrong gives 12 and −7.14 points. No positive edge can be certified. The audit also revealed a second persistent error class, revised rather than original guidance ranges, which the cleaning rules do not address. Detecting a one-point edge at the observed disagreement rate (0.105) with 80% power at two-sided $\alpha = 0.05$ requires about $8,235$ independent forecasts (McNemar approximation); clustering by issuer raises this by the design effect (about $16,470$ for a design effect of two), against 972 available.

B. Achieved skill across tasks

Table II and Fig. 2 report achieved log-loss skill relative to point-in-time base rates, for the primary nested model, with entity-clustered and time-block intervals. These are properties

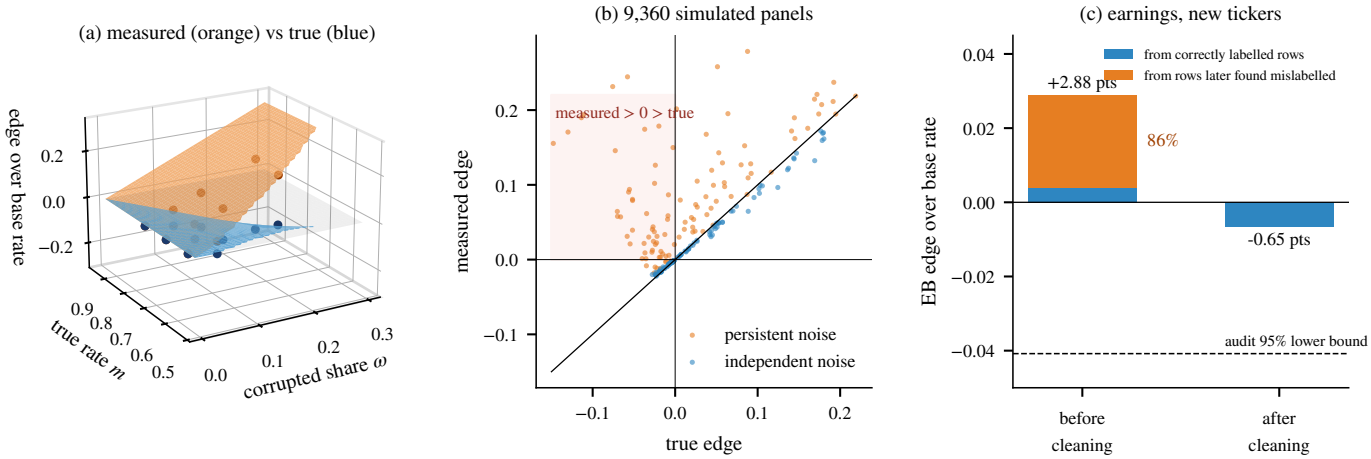


Fig. 1. Illusory edge. (a) Measured (orange) and true (blue) edge of an entity rule over the base rate under persistent corruption: surfaces from Proposition 3, points from simulation. (b) Measured against true edge in 9,360 simulated panels. (c) Earnings, new tickers: decomposition of the measured edge before cleaning, the edge after cleaning, and the audit lower bound (Clopper–Pearson version; the hypergeometric weighted bound is given in Table I).

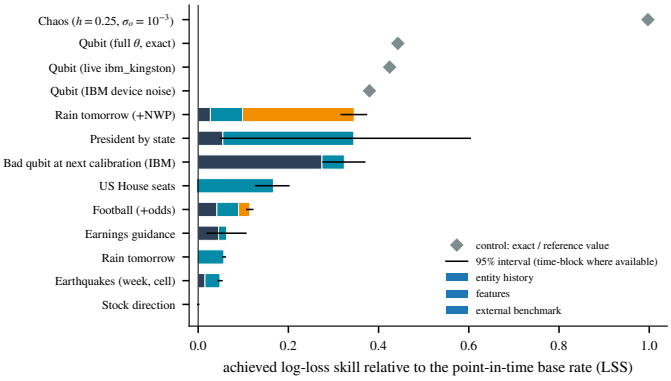


Fig. 2. Achieved log-loss skill relative to point-in-time base rates (primary nested model; stacked by achieved score improvement from entity history, features and external benchmarks; negative increments drawn to the left). Grey diamonds are reference values of controls (exact or simulated), not fitted skill.

of specified tasks, information sets and fitted systems, not of domains in general. Skill ranges from 0.346 for rain with NWP to an amount for next-day stock direction that is not distinguishable from zero under time-block resampling (−0.0007 to 0.0012); presidential-state skill is imprecise (0.050 to 0.603 across 9 elections). Some nested increments are negative (entity history for rain, House seats and stocks), which the original analysis clipped to zero; they are reported here. High accuracy is not high skill: House seats are called correctly about 90% of the time, but no model exceeds "the incumbent party holds" in accuracy (0.903 against 0.905). External benchmarks carry most of the achieved skill where they exist: NWP alone achieves 0.297, about 2.9 times the best observation-only model on the same rows, and closing odds (0.115) exceed every football model we fitted. Unexplained log loss in these tables reflects the models as well as the information; we do not call it unknowable.

C. Controls and diagnostics

On the qubit control a gradient-boosting learner reached 0.8192 (standard error 0.0012) at $n = 10^5$ against the exact $\frac{1}{2} + \frac{1}{\pi} = 0.8183$. In the Lorenz-63 control at long horizons the asymptotic Cover–Hart plug-in fell below the ensemble reference in 16 of 44 settings (14 at $h \geq 1$), so the plug-in should not be read as an upper bound at these sample sizes (Table III). The ensemble reference is itself an estimate: its sampled accuracy is not a deterministic bound on the population Bayes accuracy.

D. Exploratory: learnability–horizon curves in chaotic systems

Generative model. For each system the latent state S is drawn from the invariant measure (independent trajectories integrated with fourth-order Runge–Kutta past a burn-in, with random extra integration to decorrelate), the observation is $Z = S + \varepsilon$ with $\varepsilon \sim N(0, \mathrm{diag}(\sigma^2))$ and $\sigma = 10^{-4}$ times each coordinate's standard deviation, and the target is $\mathbf{1}\{x_1(t+h) > \text{median}\}$ at horizons $L = \lambda h \in \{0.5, \ldots, 3\}$ Lyapunov times. Training and test states are independent draws. The reference forecast $\hat{\eta}(z)$ is the fraction of 500 particles, drawn from $N(z, \mathrm{diag}(\sigma^2))$ and integrated to the horizon, that satisfy the target: a local flat-prior approximation to $p(s \mid z) \propto p(z \mid s)\mu(ds)$, centred on the observation and not on the true state. For Lorenz-63 at horizons $L \in \{0.5, 1, 2\}$ and larger noise ($\sigma \in \{0.2, 0.5\}$), where an importance-sampled posterior over 2×10^7 invariant-measure samples is feasible, the two agree in at least 0.966 of decisions and their ensemble accuracies differ by at most 0.0080; at the study's noise level, 500 against 2,000 particles changes accuracy on 1,000 test states by at most 0.000 and halving the step size leaves at least 0.996 of decisions unchanged (Appendix B). The validation covers neither the tiny-noise regime directly nor the other systems or $L = 3$; there the local approximation is assumed.

Floors. The estimated 1-NN floor $\mathbb{E}[r(1-2r)]$ is at most 0.0056 across all systems, seeds and horizons, far below the 0.05 tolerance, so the 1-NN crossings below are attainable in

TABLE II
ACHIEVED SKILL RELATIVE TO POINT-IN-TIME BASE RATES

Task	n	A_0	A	S_{acc}	LSS [entity CI]	entity	features	bench.	κ_U diag.	τ
Chaos (h=0.25, σ_o=10^{-3})	–	0.500	1.000	1.000	0.997	0.000	0.997	0.000	0.99	–
Qubit (full θ, exact)	–	0.500	0.818	0.637	0.443	0.000	0.443	0.000	–	–
Qubit (live ibm_kingston)	–	0.500	0.814	0.628	0.424	0.000	0.424	0.000	–	–
Qubit (IBM device noise)	–	0.500	0.797	0.593	0.380	0.000	0.380	0.000	–	–
Rain tomorrow (+NWP)	29,349	0.731	0.847	0.429	0.346 [0.311, 0.379]	0.027	0.071	0.248	0.52	0.99
President by state	459	0.501	0.789	0.576	0.345 [0.248, 0.454]	0.055	0.291	0.000	0.72	0.98
Bad qubit at next calibration (IBM)	60,696	0.822	0.887	0.367	0.324 [0.277, 0.369]	0.275	0.050	0.000	–	–
US House seats	6,298	0.905	0.903	-0.020	0.167 [0.139, 0.190]	-0.004	0.171	0.000	0.64	0.59
Football (+odds)	29,923	0.550	0.661	0.246	0.115 [0.102, 0.129]	0.042	0.048	0.025	0.35	0.99
Earnings guidance	972	0.660	0.668	0.023	0.063 [0.020, 0.106]	0.045	0.018	0.000	0.43	1.14
Rain tomorrow	462,762	0.742	0.758	0.062	0.057 [0.049, 0.065]	-0.001	0.058	0.000	0.51	0.96
Earthquakes (week, cell)	312,900	0.722	0.744	0.077	0.049 [0.043, 0.054]	0.015	0.034	0.000	0.49	1.01
Stock direction	483,488	0.518	0.520	0.004	0.0003 [-0.0000, 0.0005]	0.000	0.000	0.000	0.05	0.87

A_0: base-rate accuracy. LSS: achieved skill of the primary nested model; entity, features, bench.: achieved score improvements of nested models as shares of the baseline loss (signed; not mutual informations). κ_U diag.: Hellman–Raviv plug-in diagnostic from a finite-sample 1-NN error (no coverage guarantee). Time-block intervals for LSS are given in the text. Controls report exact or ensemble reference values.

TABLE III
CONTROLS: REFERENCE VALUES AND PLUG-IN DIAGNOSTICS

Setting	A^* ref.	A_{learn}	CH diag.	κ ref.	LSS	κ_U diag.
Qubit, $\sigma = 0$	0.8184	0.8162	0.8542	0.443	0.430	0.708
Qubit, $\sigma = 0.1$	0.8175	0.8166	0.8502	0.438	0.431	0.700
Qubit, $\sigma = 0.3$	0.8057	0.8056	0.8377	0.395	0.390	0.675
Qubit, $\sigma = 0.6$	0.7661	0.7670	0.7985	0.293	0.289	0.597
Qubit, $\sigma = 1$	0.7070	0.7026	0.7408	0.180	0.169	0.482
Qubit, $\sigma = 2$	0.6201	0.6200	0.6476	0.063	0.063	0.295
Lorenz, $\sigma_o = 0.001$, $h = 0.25$	1.0000	0.9880	0.9933	0.997	0.947	0.987
Lorenz, $\sigma_o = 0.001$, $h = 1$	0.9993	0.9520	0.9757	0.995	0.834	0.951
Lorenz, $\sigma_o = 0.001$, $h = 4$	0.9973	0.5600	0.6941	0.984	-0.008	0.388
Lorenz, $\sigma_o = 0.1$, $h = 1$	0.9813	0.9527	0.9690	0.940	0.819	0.938
Lorenz, $\sigma_o = 1$, $h = 1$	0.8427	0.8133	0.8851	0.542	0.463	0.770

Reference: exact (qubit) or 400-particle ensemble estimate (Lorenz). CH and κ_U: asymptotic plug-in diagnostics from finite-sample 1-NN errors, not bounds.

principle; the inconsistency of 1-NN nonetheless matters for any claim about learnability in general.

Results. For each system we drew three independent datasets and measured, for n from 10^3 to 10^6, the accuracy gap between the ensemble rule and two learners: 1-NN and k-NN with $k = \lceil\sqrt{n}\rceil$ (Fig. 3, Table IV). Crossings of the gap 0.05 are classified as interpolated between measured sample sizes, extrapolated from a learning-curve fit with a floor, unattainable, or not identifiable. Crossings already reached at the smallest n (short horizons) are excluded from the slope fits. For Lorenz-63, $\ln n^*$ is close to linear in L (mean R^2 of the exponential fit 0.999 for 1-NN and 0.996 for k-NN, against 0.976 and 0.957 for a power law in L). For Rössler the two forms fit equally well (exponential 0.960 and 0.887, power law 0.965 and 0.901), so the functional form is not identified there. The rate depends strongly on the learner: for Lorenz-63, $\gamma_L = 2.20 \pm 0.09$ for 1-NN and 3.31 ± 0.11 for k-NN (mean ± s.d. over seeds), against $D_2 \in [2.05, 2.07]$; for Rössler, 1.95 ± 0.32 and 3.31 ± 0.17 against $D_2 \in [1.82, 1.86]$. The rates also depend on the tolerance: for Lorenz-63 with 1-NN, $\gamma_L = 2.47$ at tolerance 0.02 and 2.24 at 0.10. The earlier observation that 1-NN rates are close to D_2 in three-variable flows is therefore specific to that learner and tolerance and does not describe learnability in general. For Lorenz-96 with five and six variables, fewer than three crossings per seed are interpolated within 10^6 samples at tolerance 0.05, so their rates are not identified under this protocol; at tolerance 0.10, 1-NN on Lorenz-96 with five variables gives $\gamma_L = 4.16$ against $D_2 \in [2.42, 2.47]$ (exploratory). The rates reported for Lorenz-96 in the previous version rested on a single seed, larger samples and partial extrapolation and are withdrawn. Lorenz-96 with eight variables is excluded from any headline statement. Uncertainty in λ (64 trajectories) is below 1%; D_2 varies by the range shown across scaling windows and samples.

E. Exploratory: quantum models and quantum hardware

Classical inputs. Conditionally on the training data, any quantum model applied to a classical input is a randomized function of that input, so it cannot exceed the input's Bayes ceiling (Proposition 13); it may approach that ceiling with less computation or, for structured problems, fewer samples [43], [44]. With model families selected on validation data, no quantum classifier was significantly better than the selected classical model in any of 8 domains (smallest Holm-adjusted $p = 0.576$; Table V), and the classical model was better under both dependence structures in 5. The one case that depends on the dependence model is stocks: there the variational circuit (skill +0.0002) beat the selected classical model (skill −0.0012) by −0.0010 nats under an entity bootstrap ($p = 0.0025$), but not under a month-block bootstrap ($p = 0.072$, interval $[-0.0022, +0.0005]$). Returns share market-wide shocks, so we treat the month-block result as the relevant one; both skills are close to zero. This is a failure to detect an advantage in small simulated circuits on six principal components, not an equivalence result.

Side information. In the CHSH game two players receive independent uniform bits a, b, output x, y without communicating, and win if $x \oplus y = a \wedge b$; $S = 8\omega - 4$ for win probability ω, the best noncommunicating classical strategy wins with probability 0.75, and a shared maximally entangled pair allows $\cos^2(\pi/8)$ [29], [30]. The bound 0.75 is specific to this game, not a universal classical guessing limit. In the simulated side-information experiment, the "classical" forecaster holds the

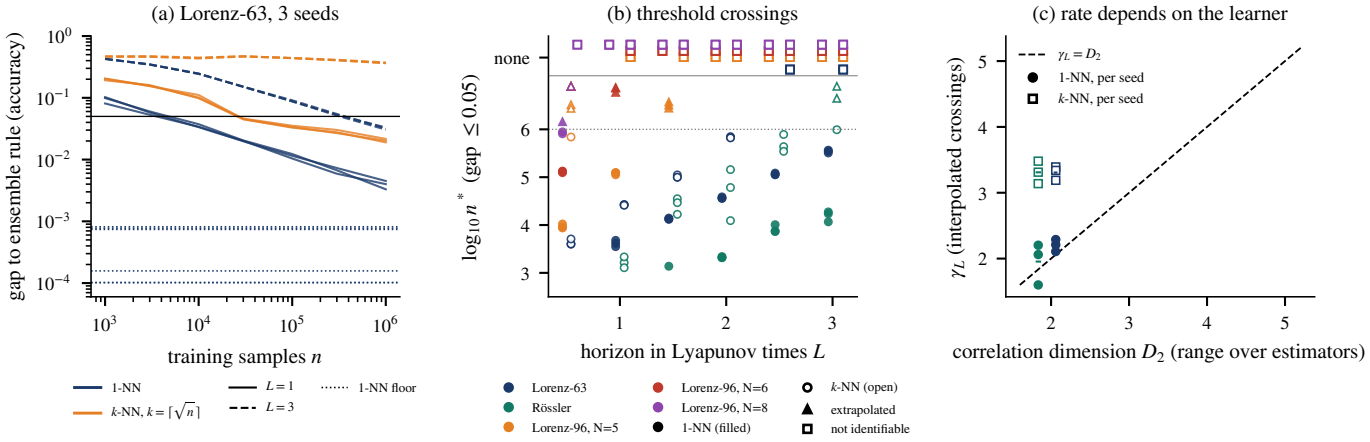


Fig. 3. Learnability–horizon curves (exploratory). (a) Lorenz-63: accuracy gap to the ensemble rule for 1-NN (blue) and k-NN (orange) at $L = 1$ and 3, three seeds each; dotted: estimated 1-NN floors. (b) Sample size at which the gap reaches 0.05, by system and learner; marker shape gives the crossing status (circle interpolated, triangle extrapolated from a floor fit and capped at $10^{6.9}$); dotted line: largest n measured; the row marked "none" collects cells with no crossing (open square: not identifiable), offset by system. (c) Per-seed growth rates from interpolated crossings against the range of correlation-dimension estimates; dashed: $\gamma_L = D_2$.

TABLE IV
LEARNABILITY–HORIZON RATES FOR TWO LEARNERS (GAP 0.05; THREE SEEDS)

System	λ	D_2 range	max 1-NN floor	γ_L, 1-NN	γ_L, k-NN	crossings 1-NN	crossings k-NN
Lorenz-63	0.9046	2.05–2.07	0.0008	2.20 ± 0.09 (3/3)	3.31 ± 0.11 (3/3)	15/0/0/0	12/0/0/6
Rössler	0.0706	1.82–1.86	0.0005	1.95 ± 0.32 (3/3)	3.31 ± 0.17 (3/3)	10/0/0/0	13/2/0/0
Lorenz-96 (N=5)	0.4773	2.42–2.47	0.0056	– (0/3)	– (0/3)	6/3/0/9	1/2/0/15
Lorenz-96 (N=6)	0.9541	3.64–3.68	0.0015	– (0/3)	– (0/3)	3/3/0/12	0/3/0/15
Lorenz-96 (N=8)‡	1.5915	4.78–4.95	0.0006	– (0/3)	– (0/3)	2/1/0/15	0/1/0/17

γ_L: mean ± s.d. over seeds of the slope of $\ln n^*$ on L using interpolated crossings only (number of seeds with ≥ 3 interpolated crossings in parentheses). Crossings: counts over 3 seeds × 6 horizons as interpolated/extrapolated/unattainable/not identifiable; the remainder of the 18 cells were already within tolerance at $n = 10^3$. ‡Exploratory only; excluded from headline statements.

TABLE V
QUANTUM AND CLASSICAL MODELS SELECTED ON VALIDATION DATA (TEST LSS)

Domain	n_{test}	selected classical	selected quantum	ΔLL (Q−C)	95% CI, entity	95% CI, time	DM p	boot. p (entity / time)	Holm p
Qubit (σ_θ=0.3)	12,000	0.397 (LR_d)	0.395 (Q1_vqc_L2)	+0.0015	[+0.0003, +0.0028]	[+0.0001, +0.0029]	0.998	0.996 / 0.980	1.000
Lorenz-63 (h=1)	12,000	0.796 (HGB_d)	0.316 (Q1_vqc_L4)	+0.3323	[+0.3272, +0.3379]	[+0.3271, +0.3374]	1.000	1.000 / 1.000	1.000
Rain	30,000	0.041 (LR_d)	0.036 (Q1_vqc_L4)	+0.0029	[+0.0003, +0.0056]	[+0.0014, +0.0044]	1.000	0.986 / 1.000	1.000
Football	10,891	0.091 (CQ1_fourier)	0.088 (Q1_vqc_L4)	+0.0021	[+0.0006, +0.0036]	[+0.0002, +0.0042]	0.986	0.995 / 0.983	1.000
US House	2,281	0.175 (CQ1_fourier)	0.171 (Q1_vqc_L4)	+0.0014	[-0.0056, +0.0083]	[-0.0048, +0.0083]	0.626	0.650 / 0.632	1.000
Earnings	293	0.008 (CQ1_fourier)	0.031 (Q1_vqc_L4)	-0.0149	[-0.0735, +0.0362]	[-0.0520, +0.0235]	0.218	0.305 / 0.224	1.000
Stocks	30,000	-0.001 (LR_d)	0.000 (Q1_vqc_L2)	-0.0010	[-0.0017, -0.0003]	[-0.0022, +0.0005]	0.073	0.003 / 0.072	0.576
Earthquakes	30,000	0.043 (CQ1_fourier)	0.037 (Q1_vqc_L2)	+0.0034	[+0.0017, +0.0052]	[+0.0024, +0.0044]	1.000	1.000 / 1.000	1.000

Selection by validation log loss within each family on identical six-dimensional inputs; the variational circuits also use the validation split for early stopping. ΔLL > 0: quantum worse. Entity: bootstrap over entities; time: bootstrap over calendar months (20 contiguous blocks for the synthetic tasks, 6 election years for the House). DM p: one-sided Diebold–Mariano (quantum better), approximate under panel dependence. Boot. p: one-sided, share of bootstrap differences ≥ 0. Holm p: Holm adjustment over the eight domains of the larger of the two bootstrap p-values.

computational-basis outcome of the partner qubit, recorded before the measurement basis is chosen; it reaches 0.751 in that protocol. A forecaster holding the partner qubit itself, and told the basis afterwards, predicts the chosen measurement perfectly (1.000); this does not reveal simultaneous outcomes of incompatible measurements [28], [45].

Live device. On 7 November 2026 we ran 36 circuits with 4,000 shots on IBM's ibm_kingston: an $R_y(\theta)$ sweep on qubits 42 and 5 (lowest reported readout errors) and the CHSH game on the coupled pair 61–62 (Fig. 4). The fitted readout errors were $e_0 = 0.0046$, $e_1 = 0.0065$ and $e_0 = 0.0043$, $e_1 = 0.0132$; the Bayes ceilings with the optimal threshold are 0.8148 (0.8132–0.8164, shot bootstrap) and 0.8127 (0.8111–0.8144), against the ideal 0.8183. The CHSH win probability was 0.8436 (standard error 0.0029). Both qubits sit on one chip, settings were fixed in the circuit, and crosstalk and communication are not excluded, so this is a demonstration of the game, not a Bell test; the device and qubits were selected by reported calibration, and the run covers one day.

Loophole-free Bell data. Reanalysing the second Delft experiment with its released selection code, we reproduce 300 trials and $S = 2.35 \pm 0.18$ [32]. With the winning condition adapted to the heralded Bell state, as in the original analysis, $\omega = 0.790$ (0.739–0.835); applying $x \oplus y = a \wedge b$ without

the state gives 0.710, which does not exceed 0.75. A one-sided binomial test against 0.75 gives $p = 0.061$; the published complete analysis [32], [33], which allows memory between trials and partially predictable settings, also reports 0.061, but the procedures differ and the agreement of the numbers does not make them equivalent. The authors' conventional analysis gives $p = 0.029$. Bob's outcome is predicted from his own setting with accuracy 0.543 and, adding Alice's setting and outcome, 0.720 (0.666–0.770; leave-one-out, identical to in-sample here).

Calibration drift. Each IBM processor is recalibrated roughly daily. Using 596, 938 distinct calibration records (repeated 30-minute polls of the same calibration were deduplicated on the calibration timestamp) of 467 qubits on three processors over 280 days [46], we forecast each qubit's next calibration on a log scale with walk-forward refits every 14 days, scoring the mean-squared-error skill against the qubit's expanding mean. Readout and two-qubit gate errors are partly forecastable one calibration ahead (skill 0.310 and 0.322), decaying to 0.148 and 0.159 two weeks ahead; T_1 and T_2 are at most weakly forecastable (0.010 [−0.015, 0.045] and 0.044 [0.029, 0.060]; replication 0.091 and 0.062), consistent with defect-driven fluctuations [34], [36]. Whether a qubit will be out of tolerance at its next calibration (readout error above the device's past 90th percentile or T_1 below its past 10th percentile; base rate 0.178) is forecast with LSS 0.324 (0.277–0.369). Intervals resample qubits and so describe generalisation across qubits of these three processors over this period, not to other processors or times; a dataset collected independently over 181 days (replication of acquisition, not an independent experiment) gives the same pattern. The primary source changed the unit of T_1 and T_2 in May 2026; a failure filter excluded the earlier records, so those series are sparse before May.

F. Preregistration scorecard

H1 has two components: H1a (the best learner within 0.02 of the qubit ceiling at $n \geq 10^4$) was met; H1b (the Cover–Hart plug-in at or above the reference in at least 80% of control settings) was not (64%). For H2, the rank correlation between predicted and measured skill was 0.89, above the preregistered 0.6, although only 4 of 7 point intervals contained the measured value and the misses were mostly toward lower skill. H3 (accuracy skill below 0.15 for earnings and stocks) and H6 (control behaviour) were consistent with the data. For H4 the simulation component was met and the earnings component (edge falling after cleaning) was observed, though the fixed-cohort analysis shows the decline reflects sample selection more than retraining. For H5 no significant quantum advantage was detected; failing to reject is not evidence of absence. The preregistered models' scores are given in the reproducibility package.

VII. Discussion

Persistent label errors. The central finding is methodological. Errors that are attached to entities and repeat over time can be learned, and time-ordered validation scores the learning as skill (Propositions 1–4). Labels extracted from filings, news and web text are prone to this structure. A disagreement audit is inexpensive: when two rules agree on most rows, a few dozen checked labels bound the measured edge on the population. In the earnings case, the audit could not certify a positive edge, and the fixed-cohort analysis attributes the apparent edge mainly to rows that should not have been scored, rather than to contaminated training.

Achieved skill versus knowability. Achieved skill mixes information with model error (Eq. (1)). Table II is a comparison of specified tasks and fitted systems; it does not establish that any outcome is intrinsically unpredictable, and near-zero stock skill in particular says nothing about the Bayes ceiling of richer information sets.

Learnability. The chaos study shows that, for nearest-neighbour learners, over horizons of 0.5–3 Lyapunov times and in the two systems where rates are identifiable, the sample size needed to approach an ensemble reference grows steeply with the horizon (close to exponentially for Lorenz-63, with the functional form unresolved for Rössler), at rates that differ between 1-NN and k-NN and with the tolerance. This is a statement about named algorithms against an approximate reference, not a law of learnability.

Quantum. Quantum resources changed what could be predicted only by changing the information available: an entangled partner's outcome. On classical data, the simulated circuits showed no detectable advantage. The more practical forecasting problem posed by quantum hardware is classical: its calibrations drift, readout errors partly predictably and coherence times largely not.

Scope. The chaos and hardware studies are exploratory and broad; each could be developed into a separate paper with designs tailored to its own questions.

VIII. Threats to Validity

Labels. Earnings labels are machine-extracted; the audit covers only disagreement rows, found one unverifiable row and a second error class (revised guidance), and the removed rows could not be relabelled. *Dependence.* Rows share shocks across entities and time; time-block intervals are reported where shared shocks are plausible but multiway dependence is not fully modelled, and the Diebold–Mariano test is approximate under panel dependence. *Data.* Unresolved point-in-time risks: revised observations (weather) and catalogues (earthquakes), survivorship in the stock universe, guidance revisions, publication delay of calibrations; House data end in 2018; the NWP benchmark is a calibrated proxy over 17 months. *Hardware.* One processor on one day for the live run, a single-chip game rather than a Bell test, third-party calibration polling, and generalisation only across qubits of three processors. *Chaos.* An approximate posterior validated only at larger noise, two learners, one relative noise level, three seeds per system, and horizons of 0.5–3 Lyapunov times. *Multiplicity.* Holm's correction is applied only within the quantum comparison. *Preregistration.* Timing rests on local records; most revision analyses are exploratory.

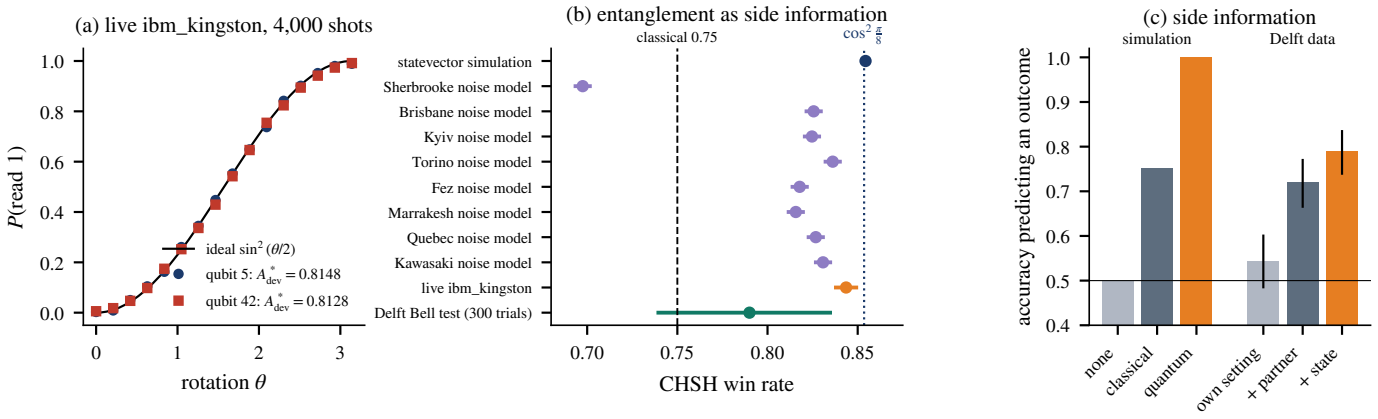


Fig. 4. Quantum hardware (exploratory). (a) Live ibm_kingston: measured $P(\text{read } 1)$ after $R_y(\theta)$ for two qubits, 4,000 shots per point, with the ideal response. (b) CHSH win probability in simulation, on eight IBM device noise models, on ibm_kingston (a single-chip game, not a Bell test) and in the Delft loophole-free Bell test (heralded-state winning condition). (c) Accuracy of predicting an outcome with and without side information.

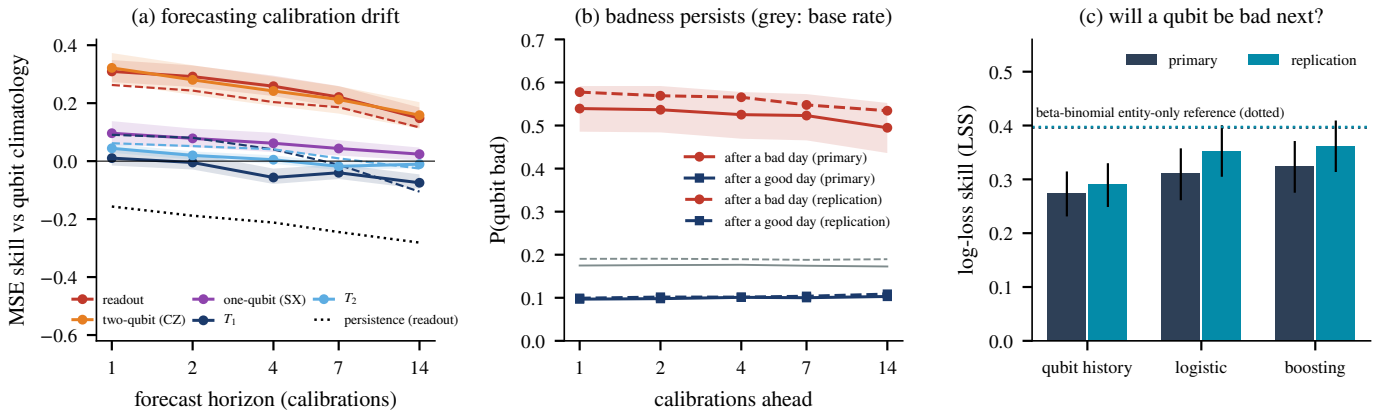


Fig. 5. Forecasting IBM processor calibrations (exploratory). (a) Mean-squared-error skill against each qubit's climatology by horizon (bands: 95% qubit-clustered intervals; dashed: replication dataset; dotted: persistence for readout error). (b) Probability that a qubit is out of tolerance k calibrations after a bad (red) or good (blue) calibration; grey: base rate. (c) Log-loss skill of three models for the out-of-tolerance task; dotted: beta-binomial entity-only reference.

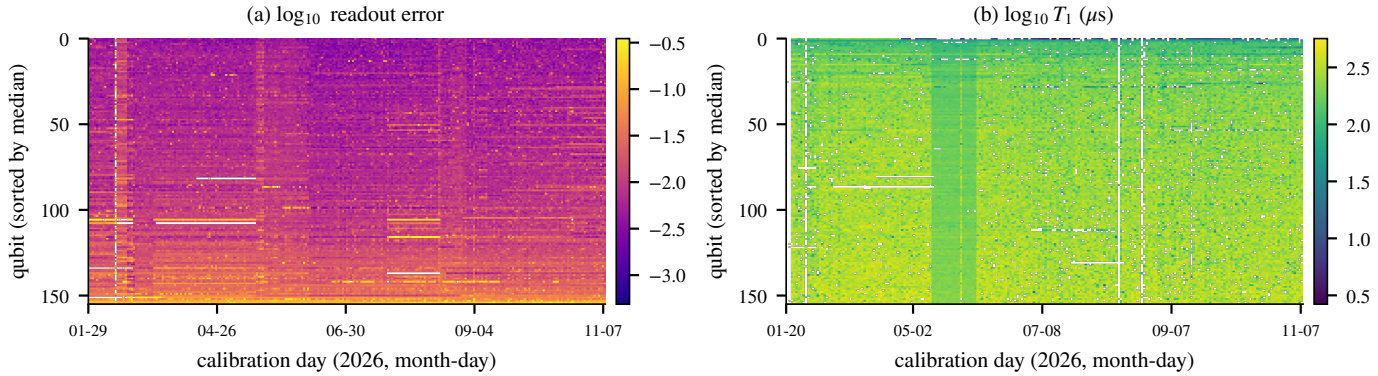


Fig. 6. Calibration histories of ibm_kingston: (a) readout error and (b) T_1 for each qubit (rows, sorted by median) at each daily calibration (columns), January–November 2026. Three-dimensional renderings and an interactive version are in the supplementary material (Appendix C).

IX. Conclusion

Label errors that persist within entities can manufacture or reverse measured edge and survive time-ordered validation; an exact identity describes the effect, and a disagreement audit with a finite-population guarantee bounds it. In a production earnings model, the apparent edge resided in rows that should not have been scored, and no positive edge could be certified. Around this core, a framework that separates Bayes ceilings of stated information sets from algorithm-specific learnability, achieved-skill comparisons across eleven tasks, and exploratory chaos and hardware studies show how much of a measured edge depends on the reference, the learner and the labels.

Appendix A
Definitions, Assumptions and Proofs

Throughout, (X, Y) is a test example drawn from a stated evaluation distribution P on $\mathcal{X}\times\{0,1\}$, where X collects every quantity available at the decision time (the information set $\mathcal{F}$ is the σ-algebra generated by X). We assume $0 < P(Y=1) < 1$, so that $H(Y) > 0$, and write $\eta(x) = P(Y = 1 \mid X = x)$ for a fixed version of the regression function, $r(x) = \min\{\eta(x), 1-\eta(x)\}$, and H_b for the binary entropy in nats. For forecasting problems the training history is part of the setting and is never part of X unless stated; temporal dependence and drift are discussed in Section V.

A. Ceilings and learnability

Definition 1 (Bayes ceiling relative to $\mathcal{F}$)**.** $A^*(\mathcal{F}) = \mathbb{E}\max\{\eta(X), 1-\eta(X)\} = 1 - \mathbb{E}\, r(X)$ and $\mathrm{LL}^*(\mathcal{F}) = H(Y \mid X) = \mathbb{E}\, H_\mathrm{b}(\eta(X))$.

Definition 2 (Algorithm-specific learnability)**.** Let $\mathcal{A}$ map a training sample $S_n = ((X_i, Y_i))_{i\le n}$ to a (possibly randomized) classifier. With $S_n \sim P^n$ independent of $(X, Y) \sim P$,

$$A_n(\mathcal{A}; P) = \mathbb{E}_{S_n} \Pr_{(X,Y),\,\xi}\{\mathcal{A}(S_n)(X;\xi) = Y\},$$
$$G_n(\mathcal{A}; P) = A^*(\mathcal{F}) - A_n(\mathcal{A}; P),$$

where ξ is internal randomness independent of (S_n, X, Y). No optimization over algorithms is implied; every finite-sample statement in this paper concerns a named algorithm. $\mathcal{A}$ is *Bayes consistent* for P if $G_n \to 0$.

Definition 3 (Knowable fraction)**.** $\kappa(\mathcal{F}) = I(Y;X)/H(Y) = 1 - H(Y \mid X)/H(Y) \in [0,1]$, the uncertainty coefficient of [6] applied to the decision-time information.

Lemma 1 (Ceiling)**.** *Let $q : \mathcal{X} \to [0,1]$ be measurable and let $\hat{Y} \mid X \sim \mathrm{Bernoulli}(q(X))$ with the randomization independent of Y given X. Then $P(\hat{Y} = Y) \le A^*(\mathcal{F})$. If $\hat{p} : \mathcal{X} \to (0,1)$ is measurable, $\mathbb{E}[-\log \hat{p}(X)^Y (1-\hat{p}(X))^{1-Y}] = H(Y \mid X) + \mathbb{E}\,\mathrm{KL}(\eta(X)\|\hat{p}(X))$.*

Proof. By conditional independence, $P(\hat{Y} = Y \mid X) = q\eta + (1-q)(1-\eta) \le \max(\eta, 1-\eta)$ almost surely; take expectations. The second identity is the decomposition of the cross-entropy given X, integrated over X. □

A classifier trained on S_n satisfies the lemma conditionally on S_n when the test example is independent of S_n; hence $A_n(\mathcal{A}; P) \le A^*(\mathcal{F})$ for every n.

Nearest neighbours. If η is continuous and X has a density on a separable metric space, the 1-NN error converges to $R_\mathrm{NN} = \mathbb{E}[2r(X)(1-r(X))]$ [23], so the asymptotic excess error of 1-NN is $\mathbb{E}[r(1-2r)] \ge 0$, with equality only if $r \in \{0, \frac{1}{2}\}$ almost surely. The k-NN rule with $k_n \to \infty$ and $k_n/n \to 0$ is universally consistent [24]. Inverting $R_\mathrm{NN} \le 2R^*(1-R^*)$ gives $R^* \ge \frac{1}{2}(1 - \sqrt{1-2R_\mathrm{NN}})$ for the *population* $R_\mathrm{NN} \le \frac{1}{2}$; substituting a finite-sample error estimate yields an asymptotic plug-in diagnostic without coverage guarantee, and it is undefined when the estimate exceeds $\frac{1}{2}$. The Hellman–Raviv inequality $R^* \le \frac{1}{2}H_2(Y \mid X)$ (bits) [47] likewise yields only a plug-in diagnostic κ_U when R^* is replaced by such an estimate.

Log-loss skill. For a forecast $\hat{p}$ and a baseline p_0 that is measurable with respect to a smaller information set $\mathcal{F}_0$,

$$\mathrm{LSS} = 1 - \frac{\mathbb{E}\ell(\hat{p})}{\mathbb{E}\ell(p_0)} = 1 - \frac{H(Y \mid X) + \mathbb{E}\,\mathrm{KL}(\eta\|\hat{p})}{H(Y \mid \mathcal{F}_0) + \mathbb{E}\,\mathrm{KL}(\eta_0\|p_0)},$$

with $\eta_0 = P(Y = 1 \mid \mathcal{F}_0)$. LSS therefore mixes the information gap $H(Y \mid \mathcal{F}_0) - H(Y \mid X)$ with the misspecification, estimation, calibration and drift losses $\mathbb{E}\,\mathrm{KL}$ of both forecasts; a small LSS does not imply a small κ, and LSS exceeds $1 - H(Y \mid X)/H(Y \mid \mathcal{F}_0)$ whenever the baseline's divergence term is large. Realized test-set averages satisfy these relations only in expectation.

B. Closed forms

Proposition 6 (Qubit)**.** *If $\theta \sim U(0,\pi)$ is known and Y is the computational-basis outcome of $R_y(\theta)|0\rangle$, then $A^* = \frac{1}{2} + \frac{1}{\pi}$, $H(Y \mid \theta) = 2\ln 2 - 1$, $H(Y) = \ln 2$ and $\kappa = 1/\ln 2 - 1$. Under the depolarizing channel $\rho \mapsto (1-p_\mathrm{d})\rho + p_\mathrm{d} I/2$, $A^* = \frac{1}{2} + (1-p_\mathrm{d})/\pi$. Under readout errors $P(\text{read } 1 \mid 0) = e_0$, $P(\text{read } 0 \mid 1) = e_1$ with $\beta = 1 - e_0 - e_1 > 0$, $a = (e_0 - e_1)/2$ and $|2a| < \beta$, the Bayes rule thresholds $p' = e_0 + \beta\sin^2(\theta/2)$ at $\frac{1}{2}$ and*

$$A^* = \tfrac{1}{2} + \tfrac{1}{\pi}\left[\beta\sin\theta_0 + a(\pi - 2\theta_0)\right], \qquad \cos\theta_0 = 2a/\beta,$$

which reduces to $\frac{1}{2} + \beta/\pi$ only when $e_0 = e_1$.

Proof. $p' - \frac{1}{2} = a - \frac{\beta}{2}\cos\theta$, and for $\theta \sim U(0,\pi)$, $\mathbb{E}|a - b\cos\theta| = \pi^{-1}[2b\sin\theta_0 + a(\pi - 2\theta_0)]$ with $b = \beta/2$, $\cos\theta_0 = a/b$, obtained by splitting the integral at θ_0. The noiseless case is $a = 0$, $\beta = 1$; depolarizing gives $a = 0$, $\beta = 1 - p_\mathrm{d}$. The entropy follows from the substitution $u = \cos\theta$: $\int_{-1}^{1} H_\mathrm{b}(\frac{1-u}{2})\,\frac{du}{\pi\sqrt{1-u^2}} = 2\ln 2 - 1$, and $\mathbb{E}p = \frac{1}{2}$. □

Proposition 7 (Beta-binomial entity model)**.** *Assume exchangeable entities with rates $p_i \sim \mathrm{Beta}(\alpha,\beta)$, known hyperparameters, and outcomes conditionally i.i.d. Bernoulli(p_i); weight entities by their number of outcomes. Then the oracle-rate accuracy is $A_1^* = \mathbb{E}[\max(p, 1-p)]$ and its conditional entropy is $\mathbb{E}[H_2(p)]$, with $p \sim \mathrm{Beta}(\alpha,\beta)$. After a history of n outcomes with k successes the predictive probability is $m_k = (\alpha + k)/(\alpha + \beta + n)$, and the finite-history Bayes accuracy is*

$$A_1^*(n) = \sum_{k=0}^{n} \binom{n}{k} \frac{B(\alpha + k, \beta + n - k)}{B(\alpha,\beta)} \max(m_k, 1 - m_k).$$

$A_1^*(n)$ is nondecreasing in n and converges to A_1^*.

Proof. The closed forms follow from Beta–Bernoulli conjugacy: the posterior after (n,k) is $\mathrm{Beta}(\alpha + k, \beta + n - k)$ and k is beta-binomial. Monotonicity holds because the history of length n is a function of the history of length $n+1$, so the Bayes accuracy of the larger σ-algebra is at least as large; convergence follows from posterior consistency for each $p_i \ne \frac{1}{2}$ and dominated convergence. □

Applied to real data, (α,β) are method-of-moments estimates and the model is an approximation; the resulting ceilings are model-based, not identified upper bounds.

C. Label noise

Let $\tilde{Y}$ be the recorded label and $\tilde{q}(x) = P(\tilde{Y} = 1 \mid X = x)$, $q = \eta$. All statements in this subsection concern *fixed* decision rules $f, g : \mathcal{X} \to \{0, 1\}$ evaluated under clean and corrupted labels; they do not describe pipelines retrained on corrupted labels unless stated.

Proposition 8 (Inflation identity)**.** $\tilde{\Delta} - \Delta = 2\,\mathbb{E}[(f(X) - g(X))(\tilde{q}(X) - q(X))]$*, where* $\Delta = P(f = Y) - P(g = Y)$ *and* $\tilde{\Delta}$ *uses* $\tilde{Y}$*.*

Proof. $P(f = \tilde{Y} \mid X) - P(f = Y \mid X) = (2f - 1)(\tilde{q} - q)$; subtract the same expression for g and integrate. □

The main-text form follows when X includes the entity and f is constant within entities, with entities weighted by their share of outcomes.

Proposition 9 (Symmetric independent noise)**.** *If* $\tilde{Y} = Y \oplus N$ *with* $N \sim \text{Bernoulli}(\rho)$*,* $\rho < \frac{1}{2}$*, independent of* (X, Y)*, then* $\tilde{\Delta} = (1 - 2\rho)\Delta$ *for fixed rules, and the Bayes rule for* $\tilde{Y}$ *equals the Bayes rule for* Y*; a Bayes-consistent learner trained on corrupted labels therefore converges to the clean Bayes rule.*

Proof. $\tilde{q} - \frac{1}{2} = (1 - 2\rho)(q - \frac{1}{2})$ and $\tilde{q} - q = \rho(1 - 2q)$; insert into Proposition 8: $\tilde{\Delta} - \Delta = -2\rho\,\mathbb{E}[(f - g)(2q - 1)] = -2\rho\Delta$. □

Proposition 10 (Sign reversal)**.** *Let* $q \equiv m > \frac{1}{2}$ *and let a fraction* ω *of entities (by outcome share) have* $\tilde{q} = 0$*, the rest* $\tilde{q} = m$*, with* $(1 - \omega)m > \frac{1}{2}$*. Let* $g \equiv 1$ *(the base-rate rule, which remains the Bayes rule for the pooled corrupted labels) and let* f *be the entity-wise Bayes rule for* $\tilde{Y}$*. Then* $\tilde{\Delta} = \omega$ *and* $\Delta = -\omega(2m - 1)$*.*

Proof. $f = 0$ on corrupted entities and 1 elsewhere; Proposition 8 gives $\tilde{\Delta} - \Delta = 2\omega m$, and directly $\Delta = \omega[(1 - m) - m]$. □

Proposition 11 (Time-ordered validation)**.** *Assume (i) for each entity, outcomes and corruption are i.i.d. over time with fixed* $(q_i, \tilde{q}_i)$ *and* $\tilde{q}_i \neq \frac{1}{2}$*; (ii) every evaluated entity has a past history whose length tends to infinity; (iii) the entity rule is the majority of the entity's past recorded labels. Then the walk-forward decisions converge almost surely to* $\mathbf{1}\{\tilde{q}_i > \frac{1}{2}\}$ *and the walk-forward measured edge converges to the in-sample measured edge. If held-out entities are forecast with the pooled base-rate rule, their entity-specific edge is zero.*

Proof. By the strong law of large numbers the empirical recorded rate converges to $\tilde{q}_i$, so the majority decision is eventually constant and equal to $\mathbf{1}\{\tilde{q}_i > \frac{1}{2}\}$ ($\tilde{q}_i \neq \frac{1}{2}$ rules out ties). Future recorded labels have the same distribution, so the expected measured accuracy of the eventual decision is the in-sample value. The last claim holds by construction of the fallback; other fallbacks (e.g. feature models) need not have zero entity-specific edge. □

Proposition 12 (Disagreement audit on a fixed population)**.** *Let a finite evaluation population of* N *rows with nonnegative weights* w_j *be fixed, together with the recorded labels and two fixed decision vectors, and let* D *be the set of* N_D *disagreement rows. (a)* $\tilde{\Delta} - \Delta$ *depends only on the wrong labels in* D*, and* $|\tilde{\Delta} - \Delta| \leq 2\sum_{j \in D_{\text{wrong}}} w_j / \sum_j w_j$*. (b) If* n *rows are drawn uniformly without replacement from* D *and* e *are found wrong, let* K_U *be the largest* K *with* $\Pr\{\text{Hypergeom}(N_D, K, n) \leq e\} > \alpha$*. With probability at least* $1 - \alpha$ *over the sampling,* $|D_{\text{wrong}}| \leq K_U$*, and therefore* $\Delta \geq \tilde{\Delta} - 2W_{(K_U)} / \sum_j w_j$*, where* $W_{(K)}$ *is the sum of the* K *largest weights in* D*; with equal weights the bound is* $\tilde{\Delta} - 2(N_D/N)(K_U/N_D)$*.*

Proof. (a) On rows outside D both rules make the same decision, so a wrong label changes both accuracies equally. A wrong label on row $j \in D$ changes the weighted difference by $\pm 2w_j / \sum w$. (b) For fixed K the number of wrong rows in the sample is hypergeometric; the set $\{K : \Pr(X \leq e \mid K) > \alpha\}$ is a one-sided $(1 - \alpha)$ confidence set by the usual test inversion, since $\Pr(X \leq e \mid K)$ is nonincreasing in K. The weight bound is the worst case over which K_U rows are wrong. □

The statement concerns measured edge on the fixed population as labelled; it says nothing about future predictive edge, which requires a separate sampling model for future cases. The binomial Clopper–Pearson bound is valid for sampling with replacement and conservative relative to (b) for sampling without replacement.

D. Classical inputs and quantum models

Proposition 13 (Classical-input ceiling)**.** *Let a model with parameters* $\phi = \phi(S_n)$ *fitted on training data encode a test input* x *into a state* $\rho_\phi(x)$*, apply a measurement, and output* $\hat{Y}$*. If the test example is independent of* S_n *and the measurement randomness is independent of* Y *given* (X, S_n)*, then conditionally on* S_n *the chain* $Y \to X \to \hat{Y}$ *is Markov,* $I(Y; \hat{Y} \mid S_n) \leq I(Y; X)$ *and* $P(\hat{Y} = Y) \leq A^*(\mathcal{F})$*.*

Proof. Given S_n, $\hat{Y}$ is a randomized function of X; apply Lemma 1 conditionally and the data-processing inequality, then average over S_n. □

The proposition excludes accuracy above the Bayes ceiling of the classical input; it permits a quantum model to approach that ceiling with less computation or, for structured η, fewer samples [43], [44]. It does not apply when the predictor holds a quantum system correlated with the predicted one, which is a different information set.

Appendix B
Reproducibility

The preregistration's SHA-256 digest is
`abee64cc5cd31d0b2212550d03508580`
`561aa6cc47eb9462462ddb88a983d33c`.
The repository (git branch `revision-2`; original analyses on `master`) contains the shared library with unit tests (closed forms, inflation identity, 1-NN asymptotics, audit coverage, GPU simulator against Qiskit), one builder per task, the walk-forward evaluator, the chaos study (`run_law3.py`) and posterior validation (`validate_posterior.py`), the revision analyses (`rev2_analyses.py`), the validation-selected quantum comparison (`run_quantum2.py`), a feature-and-timestamp specification with the leakage audit, an environment lockfile, file hashes, and a manifest mapping every table and figure to its script. Raw data whose licences

forbid redistribution are replaced by acquisition scripts and hashes. Data sources: Meteostat (CC BY-NC 4.0), Open-Meteo, football-data.co.uk, MIT Election Data and Science Lab (CC0), Yahoo Finance via `yfinance`, USGS ComCat, SEC EDGAR, IBM calibration histories (CC BY 4.0; replication dataset without licence, analysed but not redistributed), the Delft Bell-test data (4TU.ResearchData), IBM device noise models in `qiskit-ibm-runtime`, and IBM Quantum job `dbnk4cc4148s73915tpg`; all accessed in November 2026.

Appendix C
Three-Dimensional Renderings

Figure 7 reproduces renderings from the interactive companion: the calibration surfaces behind Fig. 6, ensemble certainty on the Lorenz attractor, and the original single-seed learnability surface. They are visual summaries of results reported above and support no additional inference; the learnability surface comes from the superseded single-seed protocol.

Acknowledgment

Computational experiments, analysis code and drafting were assisted by an AI system (Claude, Anthropic), which is not an author. All reported numbers are produced by the released code.

References

[1] W. Ciconte, M. Kirk, and J. W. Tucker, "Does the midpoint of range earnings forecasts represent managers' expectations?" *Rev. Account. Stud.*, vol. 19, no. 2, pp. 628–660, 2014.

[2] B. Frénay and M. Verleysen, "Classification in the presence of label noise: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 25, no. 5, pp. 845–869, 2014.

[3] X. Xia, T. Liu, B. Han, N. Wang *et al.*, "Part-dependent label noise: Towards instance-dependent label noise," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020, arXiv:2006.07836.

[4] M. Bernhardt, D. C. Castro, R. Tanno, A. Schwaighofer *et al.*, "Active label cleaning for improved dataset quality under resource constraints," *Nature Communications*, vol. 13, no. 1, p. 1161, 2022.

[5] J. Kossen, S. Farquhar, Y. Gal, and T. Rainforth, "Active testing: Sample-efficient model evaluation," in *Proc. ICML (PMLR 139)*, 2021, arXiv:2103.05331.

[6] H. Theil, "On the estimation of relationships involving qualitative variables," *American Journal of Sociology*, vol. 76, no. 1, pp. 103–154, 1970.

[7] S. V. Weijs, R. van Nooijen, and N. van de Giesen, "Kullback–leibler divergence as a forecast skill score with classic reliability–resolution–uncertainty decomposition," *Monthly Weather Review*, vol. 138, no. 9, pp. 3387–3399, 2010.

[8] M. S. Roulston and L. A. Smith, "Evaluating probabilistic forecasts using information theory," *Monthly Weather Review*, vol. 130, no. 6, pp. 1653–1660, 2002.

[9] H. Song, M. Kim, D. Park, Y. Shin, and J.-G. Lee, "Learning from noisy labels with deep neural networks: a survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8135–8153, 2023.

[10] N. Natarajan, I. S. Dhillon, P. Ravikumar, and A. Tewari, "Learning with noisy labels," in *Advances in Neural Information Processing Systems 26 (NIPS)*, 2013, pp. 1196–1204.

[11] C. Northcutt, L. Jiang, and I. Chuang, "Confident learning: estimating uncertainty in dataset labels," *J. Artif. Intell. Res.*, vol. 70, pp. 1373–1411, 2021.

[12] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021, arXiv:2103.14749.

[13] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[14] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[15] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *J. Amer. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007.

[16] C. Song, Z. Qu, N. Blumm, and A.-L. Barabási, "Limits of predictability in human mobility," *Science*, vol. 327, no. 5968, pp. 1018–1021, 2010.

[17] T. Martin, J. M. Hofman, A. Sharma, A. Anderson, and D. J. Watts, "Exploring limits to prediction in complex social systems," in *Proc. 25th Int. Conf. World Wide Web (WWW)*, 2016, pp. 683–694.

[18] J. M. Hofman, A. Sharma, and D. J. Watts, "Prediction and explanation in social systems," *Science*, vol. 355, no. 6324, pp. 486–488, 2017.

[19] M. J. Salganik, I. Lundberg, A. T. Kindel *et al.*, "Measuring the predictability of life outcomes with a scientific mass collaboration," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 117, no. 15, pp. 8398–8403, 2020.

[20] E. N. Lorenz, "The predictability of a flow which possesses many scales of motion," *Tellus*, vol. 21, no. 3, pp. 289–307, 1969.

[21] F. Zhang, Y. Q. Sun, L. Magnusson, R. Buizza, S.-J. Lin, J.-H. Chen, and K. Emanuel, "What is the predictability limit of midlatitude weather?" *J. Atmos. Sci.*, vol. 76, no. 4, pp. 1077–1091, 2019.

[22] P. Bauer, A. Thorpe, and G. Brunet, "The quiet revolution of numerical weather prediction," *Nature*, vol. 525, no. 7567, pp. 47–55, 2015.

[23] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. 13, no. 1, pp. 21–27, 1967.

[24] C. J. Stone, "Consistent nonparametric regression," *The Annals of Statistics*, vol. 5, no. 4, pp. 595–620, 1977.

[25] J. D. Farmer and J. J. Sidorowich, "Predicting chaotic time series," *Physical Review Letters*, vol. 59, no. 8, pp. 845–848, 1987.

[26] M. Casdagli, "Nonlinear prediction of chaotic time series," *Physica D*, vol. 35, no. 3, pp. 335–356, 1989.

[27] J. Pathak, B. Hunt, M. Girvan, Z. Lu, and E. Ott, "Model-free prediction of large spatiotemporally chaotic systems from data: A reservoir computing approach," *Physical Review Letters*, vol. 120, no. 2, p. 024102, 2018.

[28] R. König, R. Renner, and C. Schaffner, "The operational meaning of min- and max-entropy," *IEEE Trans. Inf. Theory*, vol. 55, no. 9, pp. 4337–4347, 2009.

[29] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.*, vol. 23, no. 15, pp. 880–884, 1969.

[30] B. S. Cirel'son, "Quantum generalizations of Bell's inequality," *Lett. Math. Phys.*, vol. 4, no. 2, pp. 93–100, 1980.

[31] B. Hensen, H. Bernien, A. E. Dréau, A. Reiserer, N. Kalb, M. S. Blok, J. Ruitenberg, R. F. L. Vermeulen, R. N. Schouten, C. Abellán, W. Amaya, V. Pruneri, M. W. Mitchell, M. Markham, D. J. Twitchen, D. Elkouss, S. Wehner, T. H. Taminiau, and R. Hanson, "Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres," *Nature*, vol. 526, no. 7575, pp. 682–686, 2015.

[32] B. Hensen, N. Kalb, M. S. Blok, A. E. Dréau, A. Reiserer, R. F. L. Vermeulen, R. N. Schouten, M. Markham, D. J. Twitchen, K. Goodenough, D. Elkouss, S. Wehner, T. H. Taminiau, and R. Hanson, "Loophole-free Bell test using electron spins in diamond: second experiment and additional analysis," *Sci. Rep.*, vol. 6, p. 30289, 2016.

[33] D. Elkouss and S. Wehner, "(nearly) optimal P values for all Bell inequalities," *npj Quantum Information*, vol. 2, no. 1, p. 16026, 2016.

[34] P. V. Klimov, J. Kelly, Z. Chen, M. Neeley, A. Megrant, B. Burkett, R. Barends *et al.*, "Fluctuations of energy-relaxation times in superconducting qubits," *Phys. Rev. Lett.*, vol. 121, no. 9, p. 090502, 2018.

[35] J. J. Burnett, A. Bengtsson, M. Scigliuzzo, D. Niepce, M. Kudra, P. Delsing, and J. Bylander, "Decoherence benchmarking of superconducting qubits," *npj Quantum Inf.*, vol. 5, p. 54, 2019.

[36] M. Carroll, S. Rosenblatt, P. Jurcevic, I. Lauer, and A. Kandala, "Dynamics of superconducting qubit relaxation times," *npj Quantum Information*, vol. 8, no. 1, p. 132, 2022.

[37] T. Proctor, M. Revelle, E. Nielsen, K. Rudinger *et al.*, "Detecting and tracking drift in quantum information processors," *Nature Communications*, vol. 11, no. 1, p. 5396, 2020.

[38] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[39] H. Robbins, "An empirical Bayes approach to statistics," in *Proc. 3rd Berkeley Symp. Math. Statist. Probab.*, vol. 1. Univ. California Press, 1956, pp. 157–163.

[40] B. Efron and C. Morris, "Data analysis using Stein's estimator and its generalizations," *J. Amer. Statist. Assoc.*, vol. 70, no. 350, pp. 311–319, 1975.

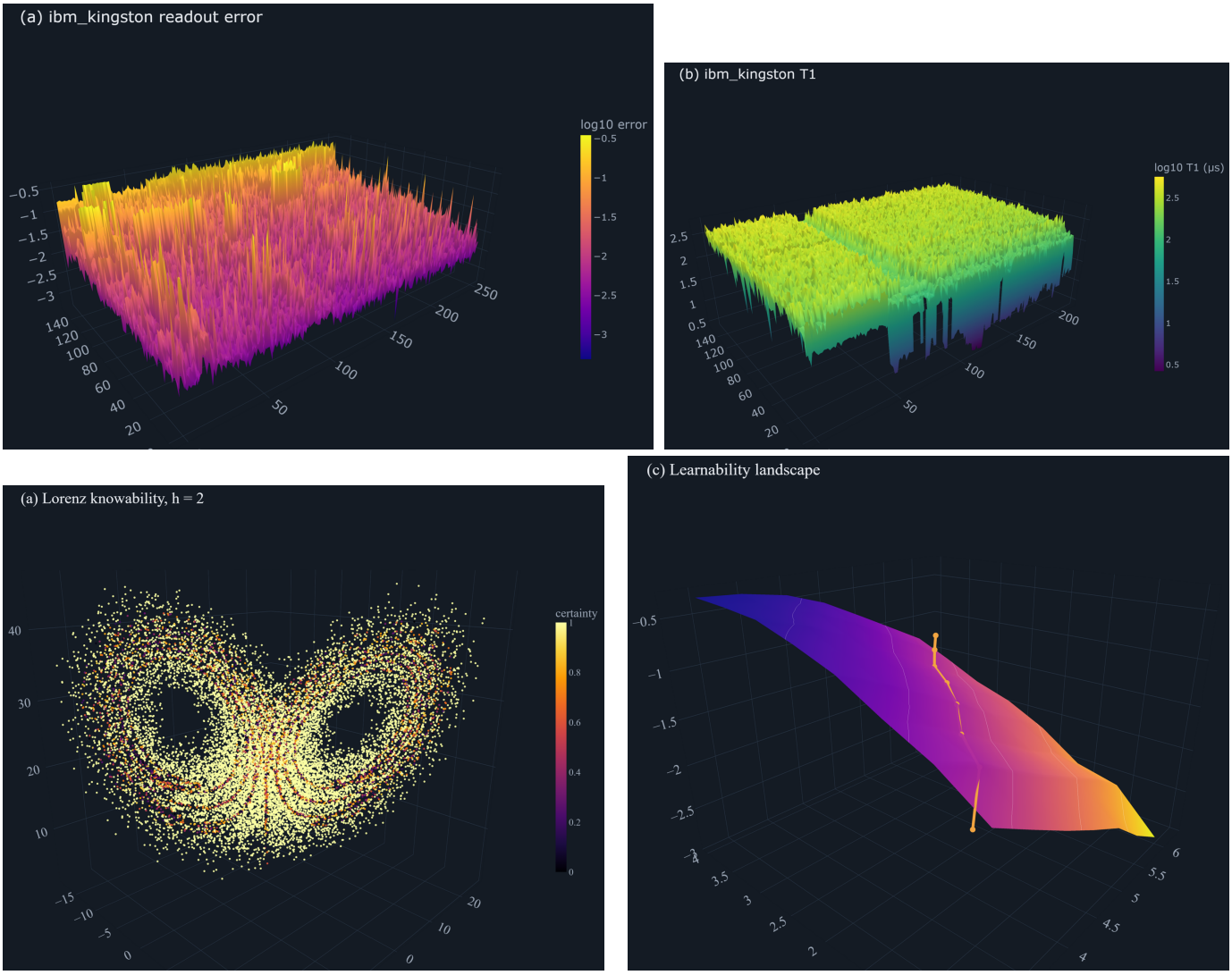


Fig. 7. Renderings from the interactive companion. Top: ibm_kingston calibration surfaces of Fig. 6, $\log_{10}$ readout error and $\log_{10} T_1$ (μs) over qubit (depth, sorted by median) and calibration day (horizontal). Bottom left: ensemble certainty $2|\hat{\eta} - \frac{1}{2}|$ of the sign of $x(t+2)$ on the Lorenz attractor ($\sigma_o = 0.1$, original single-seed study). Bottom right: the original single-seed 1-NN learnability surface for Lorenz-63 ($\log_{10}$ gap over $\log_{10} n$ and h), superseded by Fig. 3.

[41] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *J. Bus. Econ. Statist.*, vol. 13, no. 3, pp. 253–263, 1995.
[42] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.
[43] Y. Liu, S. Arunachalam, and K. Temme, "A rigorous and robust quantum speed-up in supervised machine learning," *Nature Phys.*, vol. 17, no. 9, pp. 1013–1017, 2021.
[44] H.-Y. Huang, R. Kueng, and J. Preskill, "Information-theoretic bounds on quantum advantage in machine learning," *Phys. Rev. Lett.*, vol. 126, no. 19, p. 190505, 2021.
[45] M. Berta, M. Christandl, R. Colbeck, J. M. Renes, and R. Renner, "The uncertainty principle in the presence of quantum memory," *Nature Phys.*, vol. 6, no. 9, pp. 659–662, 2010.
[46] C. C. Norton, "qiskit-calibration-drift: IBM quantum processor calibration time series," Hugging Face dataset phanerozoic/qiskit-calibration-drift (CC BY 4.0), 2026, accessed Nov. 2026.
[47] M. E. Hellman and J. Raviv, "Probability of error, equivocation, and the Chernoff bound," *IEEE Trans. Inf. Theory*, vol. 16, no. 4, pp. 368–372, 1970.